

1. Notes

a. Results here should not be compared directly to the historical analysis in other ppt slides (by Emircan). Key differences:

This report's slides (model results)

- USFS-**Owned** resources
- Resource uses for **wildfires** only

Emircan's slides (historical statistics)

- USFS-**Managed** resources
- Resource uses may include **wildfires, prescribed fires, and others.**

b. Two-stage optimization approach:

- Stage 1: minimize the average annual shortage counts of all resources (**equal weight for lacking each unit of every resource**)
- Stage 2: fix stage 1's optimal shortage, additionally balance workload & minimize resource mobilization across GACCs

Example: same 1509 annual shortage in solutions from both stages, but:

Stage 1 – mobilization – engine 3

Engine_3	AICC	NWCC	ONCC	OSCC	NRCC	GBCC	SWCC	RMCC	EACC	SACC
AICC	0	0	0	0	0	0	0	0	0	0
NWCC	0	1214	2	11	5	4	1	6	9	13
ONCC	0	9	3046	81	53	142	14	14	27	41
OSCC	0	17	61	3815	21	88	46	28	22	67
NRCC	0	0	0	0	0	0	0	0	0	0
GBCC	0	0	0	0	0	290	0	0	0	0
SWCC	0	0	2	7	1	2	1084	1	2	6
RMCC	0	0	0	0	0	0	0	109	0	0
EACC	0	0	0	0	0	0	0	0	30	0
SACC	0	0	0	0	0	0	0	0	0	0

Stage 2 – mobilization – engine 3

Engine_3	AICC	NWCC	ONCC	OSCC	NRCC	GBCC	SWCC	RMCC	EACC	SACC
AICC	0	0	0	0	0	0	0	0	0	0
NWCC	1	552	0	0	26	0	0	1	1	0
ONCC	0	0	2830	0	25	3	0	1	0	0
OSCC	0	0	3	3762	0	23	0	5	1	1
NRCC	0	0	0	0	0	0	0	0	0	0
GBCC	0	0	0	0	0	387	0	0	0	0
SWCC	0	0	0	0	0	0	654	1	2	15
RMCC	0	0	0	0	0	0	0	61	0	0
EACC	0	0	0	0	0	0	0	0	13	0
SACC	0	0	0	0	0	0	0	0	0	0

Resource type	Stage 1 Workload (days)	Stage 2 Workload (days)
Crew 1	60	69
Crew 2	13	21
Crew 2IA	41	23
Dozer 1		
Dozer 2	24	21
Dozer 3	11	20
Dozer 4		
Engine 3	25	20
Engine 4	19	21
Engine 5		
Engine 6	9	13
Engine 7	3	8
Module 1		
Module 2	20	27
Water 1	16	21
Water 2	16	16

2. Scenarios

a. Resource staffing levels:

- 6 years of the data 2020-2025. Every resource ID that is used for at least one wildfire is extracted.
- For each year and for each resource, find the day of maximum usage to identify staffing level for that resource in that year.
- Averaging the above yearly results to use in models. We want to see how the model respond to historical fire situations.

	AICC	NWCC	ONCC	OSCC	NRCC	GBCC	SWCC	RMCC	EACC	SACC
Crew_1	0	10	14	20	6	6	14	5	1	2
Crew_2IA	1	15	7	9	9	7	4	4	3	2
Crew_2	0	2	3	12	1	1	0	0	0	1
Engine_3	0	29	128	184	0	16	30	4	1	0
Engine_4	0	17	0	4	9	27	2	3	0	0
Engine_5	0	0	0	0	0	0	0	0	0	0
Engine_6	2	68	3	11	70	39	68	49	39	53
Engine_7	0	2	3	15	0	5	0	2	0	4
Dozer_1	0	0	0	0	0	0	0	0	0	0
Dozer_2	0	1	13	9	0	0	8	3	0	3
Dozer_3	0	1	0	0	0	0	1	2	10	11
Dozer_4	0	0	0	0	0	0	0	0	1	5
Module_1	0	1	1	0	1	3	0	2	1	0
Module_2	0	4	3	4	3	5	2	4	1	0
Water_1	0	1	1	0	0	0	1	0	1	0
Water_2	0	1	22	24	3	0	2	0	0	0

Total
1205 units

5 alternatives: 100%, 90%, 80%, 70%, to 60% (all numbers in the above table decrease to 60%, such as 10 becomes 6), respectively associated with a total of 1205, 1085, 964, 844, and 723 resource units.

2. Scenarios

b. Resource substitution rules: e.g. if demand is crew type 2, supply can be type 1, type 2IA, or type 2.

- Crew 1 > Crew 2IA > Crew 2
- Engine 3 > Engine 4 > Engine 5 > Engine 6 > Engine 7
- Dozer 1 > Dozer 2 > Dozer 3 > Dozer 4
- Module 1 > Module 2
- Water tender 1 > Water tender 2

c, 4 GACC's risk-tolerance levels (no, low, medium, high risk)

- Constant sharing levels if GACC PL (GPL) is 1 or 2, less sharing if $GPL > 2$ and $GPL \leq NPL$ (National PL), No sharing if $GPL = 5$.
- GACCs must make resources available to send out as below, but other GACCs may not use up those availabilities.

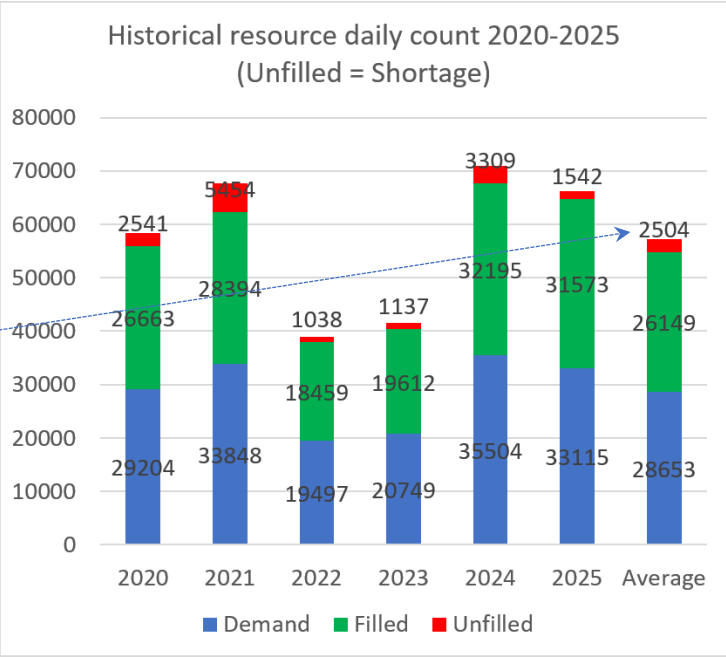
No risk (no sharing)						Low risk, share up to 25%						Medium risk, share up to 50%						High risk, share up to 75%					
NPL→ GPL↓	1	2	3	4	5	NPL→ GPL↓	1	2	3	4	5	NPL→ GPL↓	1	2	3	4	5	NPL→ GPL↓	1	2	3	4	5
1	0	0	0	0	0	1	25	25	25	25	25	1	50	50	50	50	50	1	75	75	75	75	75
2	0	0	0	0	0	2	25	25	25	25	25	2	50	50	50	50	50	2	75	75	75	75	75
3	0	0	0	0	0	3	0	0	12	12	12	3	0	0	25	25	25	3	0	0	32	32	32
4	0	0	0	0	0	4	0	0	0	12	12	4	0	0	0	25	25	4	0	0	0	32	32
5	0	0	0	0	0	5	0	0	0	0	0	5	0	0	0	0	0	5	0	0	0	0	0

- PSW rule is applied for 50% mobilization cut during non-peak season: all above numbers are for peak season (days 121 to 334 or May 1st to Nov 30th), and all those numbers decrease to 50% for non-peak season (such as 12% becomes 6% for day 100).
- The “Medium risk” case mimics well some GACC plans for sharing when GPL is 1 or 2. But for $GPL = 3, 4, 5$, GACCs often emphasizes that they are heavy importers. Here, we still force GACCs to share at GPL 3 and 4 and NPL is same or higher

3. Results (by stage-2 solutions) vs Historical Data

a, **Minimum annual shortage from optimal model solutions**

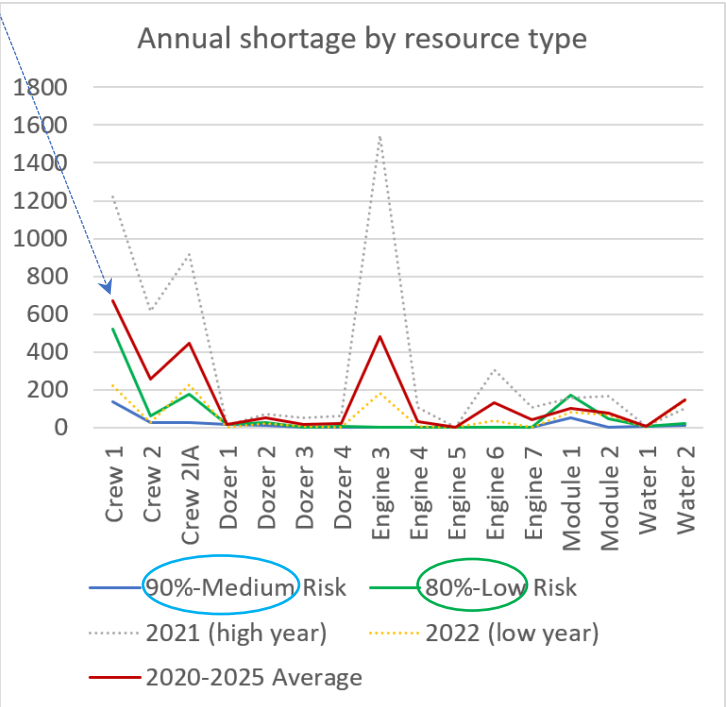
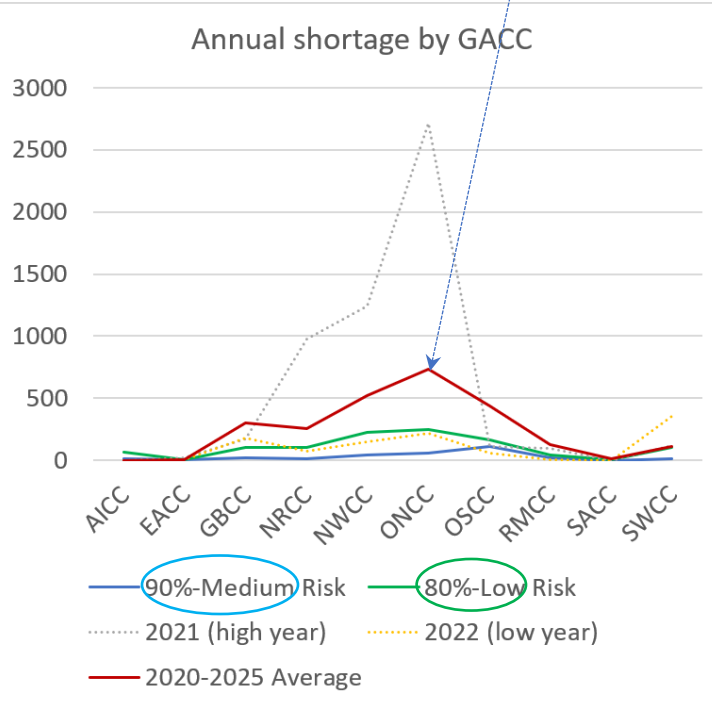
Risk-tolerant level → Staffing levels↓	No risk (share 0%)	Low risk (share up to 25%)	Medium risk (share up to 50%)	High risk (share up to 75%)	Historical average (2020-2025)
100% (1205 units)	1855.1	533.9	225.3	184.4	2504.5
90% (1085 units)	2107.6	692.6	290.8	227.4	
80% (964 units)	2479.0	1064.4	435.9	351.8	
70% (844 units)	2928.1	1466.8	745.3	605.6	
60% (723 units)	3518.8	2075.3	1188.6	995.6	



Trend:

- Shortage decreases when GACCs share more
- Shortage decreases when staffing levels are higher

Model performs well compared to historical data.



3. Results (by stage-2 solutions) vs Historical Data

b, Balanced workload (days on assignments annually) from optimal model solutions

Model	100% staff Low Risk	90% staff Low Risk	80% staff Low Risk	70% staff Low Risk	60% staff Low Risk	100% staff Med Risk	90% staff Med Risk	80% staff Med Risk	70% staff Med Risk	60% staff Med Risk
Crew 1	81	89	95	105	113	75	84	95	111	125
Crew 2	21	23	26	30	35	21	23	26	30	35
Crew 2IA	35	35	40	43	52	46	46	50	49	55
Dozer 1										
Dozer 2	23	25	30	32	36	23	25	30	32	36
Dozer 3	12	13	14	17	19	12	13	14	17	19
Dozer 4	8	8	10	10	9	9	9	10	10	12
Engine 3	21	23	26	30	35	21	23	26	30	35
Engine 4	21	23	26	30	35	21	23	26	30	35
Engine 5										
Engine 6	15	17	19	21	25	15	17	19	21	25
Engine 7	8	8	9	11	13	8	8	9	11	13
Module 1	35	35	40	43	45	46	46	50	49	51
Module 2	33	33	40	41	45	33	33	43	45	54
Water 1	21	23	26	30	35	21	23	26	30	35
Water 2	19	20	22	26	29	19	20	23	26	29
Shortage	534	693	1064	1467	2075	225	291	436	745	1189

Historical	2020	2021	2022	2023	2024	2025	Average
Crew 1	55	57	57	42	64	61	56
Crew 2	40	34	38	22	53	41	38
Crew 2IA	43	47	44	43	55	51	47
Dozer 1							
Dozer 2	24	39	11	13	22	24	22
Dozer 3	3	9	6	12	12	14	9
Dozer 4	0	0	0	10	7	11	5
Engine 3	21	21	14	14	24	24	20
Engine 4	18	17	10	12	19	17	15
Engine 5							
Engine 6	15	16	10	13	19	19	15
Engine 7	3	6	2	8	9	6	6
Module 1	35	42	35	28	43	46	38
Module 2	26	40	24	28	39	29	31
Water 1	4	4	4	5	1	1	3
Water 2	15	22	13	11	22	21	17
Shortage	2541	5454	1038	1137	3309	1542	2504

- Blue annual shortages (see previous slide) are added here for reference. All black numbers presented here use the same unit of “working days per year”
- 3 green columns are somewhat comparable because they both use 100% the average annual staffing levels from 6 historical years (2020-2025). It seems workload from model results align well with historical workload.
- One note that: the staffing levels used in our model to deal with wildfires only, but those same staffing levels are used historically for multiple purposes: wildfires, prescribed fires, prepositions, and others. This raises a question:

Should we and can we integrate demands from prescribed fires... into our models? (Erin may know best about the challenge of data processing)

3. Results (by stage-2 solutions) vs Historical Data

c, **Resource mobilization:** cherry pick model with 100% staffing levels and GACCs' medium risk-tolerant level (share up to 50% resources)

Peak-season Mobilization rate (percentage of staffing levels) a GACC has to make available to support the other GACCs

For non-peak season (days outside of May-November), the below percentages decrease to half (50%) of the presented values

NPL													NPL													NPL																																						
1													2													3													4													5												
GPL	Row Labels	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5																		
Crew 1																																																																
	AICC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0																
	EACC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0															
	GBCC	50	50	0	0	0	50	0	0	0	0	50	50	0	0	0	50	50	16	0	0	50	0	0	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0														
	NRCC	50	50	0	0	0	50	0	0	0	0	50	0	16	0	0	50	50	0	0	0	0	0	0	0	16	16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0														
	NWCC	50	50	0	0	0	50	50	0	0	0	0	50	0	0	0	0	0	0	0	0	0	0	0	50	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0														
	ONCC	50	50	0	0	0	50	0	0	0	0	50	50	21	0	0	0	50	21	0	0	0	50	0	0	21	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0														
	OSCC	50	50	0	0	0	50	50	0	0	0	50	50	0	0	0	0	0	0	0	0	0	0	0	0	25	25	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0														
	RMCC	40	40	0	0	0	40	40	0	0	0	40	40	0	0	0	40	20	20	0	0	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0														
	SACC	50	50	0	0	0	50	50	0	0	0	50	50	0	0	0	50	0	0	0	0	0	50	50	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0														
	SWCC	50	50	0	0	0	50	50	0	0	0	50	50	21	0	0	50	0	0	0	0	0	50	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0														
Crew 2																																																																
	AICC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	EACC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	GBCC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	NRCC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	NWCC	50	50	0	0	0	0	50	0	0	0	0	50	0	0	0	0	0	0	0	0	0	50	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	ONCC	33	33	0	0	0	0	0	0	0	0	33	33	0	0	0	33	0	0	0	0	0	33	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	OSCC	50	50	0	0	0	0	50	0	0	0	50	50	0	0	0	0	50	0	0	0	50	50	25	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	RMCC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	SACC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	SWCC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
Crew 2IA																																																																
	AICC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	EACC	33	0	0	0	0	0	33	0	0	0	0	33	0	0	0	0	33	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	GBCC	42	42	0	0	0	42	0	0	0	0	42	42	0	0	0	42	42	14	0	0	0	42	14	14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	NRCC	44	44	0	0	0	44	44	0	0	0	44	0	22	0	0	0	44	0	0	0	44	22	22	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	NWCC	46	46	0	0	0	46	46	0	0	0	46	20	0	0	0	46	46	20	0	0	0	46	0	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	ONCC	42	42	0	0	0	42	0	0	0	0	42	42	0	0	0	0	0	0	0	0	42	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	OSCC	44	44	0	0	0	44	44	0	0	0	44	44	0	0	0	44	0	0	22	0	0	22	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	RMCC	50	50	0	0	0	50	0	0	0	0	50	50	25	0	0	0	0	0	0	0	50	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	SACC	50	50	0	0	0	50	0	0	0	0	50	50	0	0	0	0	0	0	0	0	50	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	SWCC	50	50	0	0	0	50	50	0	0	0	50	50	25	0	0	50	0	0	25	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
Dozer 2																																																																
	AICC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	EACC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	GBCC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	NRCC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	NWCC	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	ONCC	46	46	0	0	0	46	0	0	0	0	46	46	23	0	0	0	46	0	0	0	46	0	0	23	23	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	OSCC	44	44	0	0	0	44	0	0	0	0	44	44	22	0	0	0	0	0	0	0	0	0	0	0	22	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0													
	RMCC	33	33	0	0	0	33	33	0	0	0	33	33	0	0	0	33	0	0	0	0	0	33	0	0	0	0	0	0																																			

Maximum staffing level (100%) = Mobilization rate + Drawdown rate

3. Results (by stage-2 solutions) vs Historical Data

c, **Resource mobilization:** cherry pick model with 100% staffing levels and GACCs' medium risk-tolerant level (share up to 50% resources)

Model 100% - Medium risk

GACCs	Demand	Shortage	Supply	Internal supply	External supply
AICC	202	11	191	17	174
EACC	495	2	493	474	20
GBCC	3201	15	3186	2909	278
NRCC	2185	9	2176	1990	186
NWCC	4211	30	4181	3913	268
ONCC	5323	36	5287	4700	587
OSCC	7321	91	7229	7191	39
RMCC	1342	22	1319	1247	72
SACC	871	0	871	837	34
SWCC	3502	10	3493	3238	254
Total	28653	226	28427	26515	1912

Historical (average 2020-2025)

GACCs	Demand	Shortage	Supply	Internal supply	External supply
AICC	202	1	201	9	193
EACC	495	7	488	390	98
GBCC	3201	298	2904	2158	746
NRCC	2186	259	1927	1291	636
NWCC	4211	523	3688	2906	782
ONCC	5323	735	4589	3052	1537
OSCC	7321	435	6886	6205	681
RMCC	1342	127	1215	746	469
SACC	871	11	860	595	265
SWCC	3502	110	3392	2240	1152
Total	28653	2504	26149	19591	6558

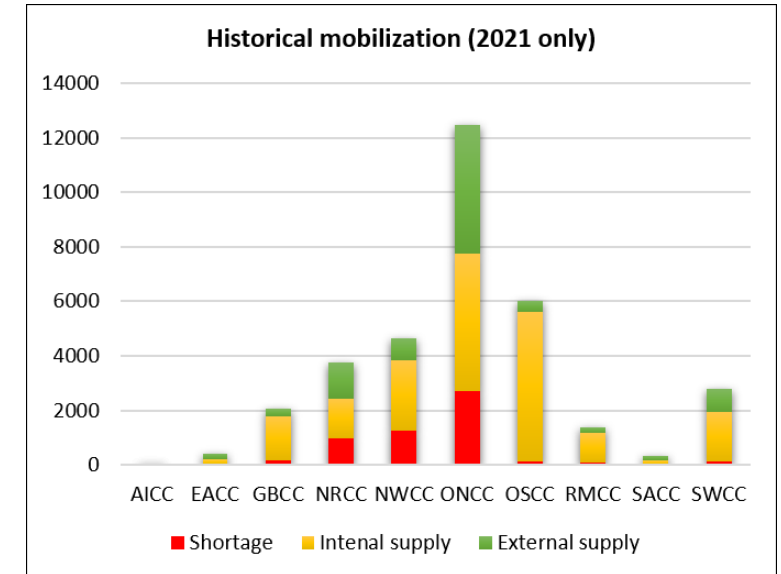
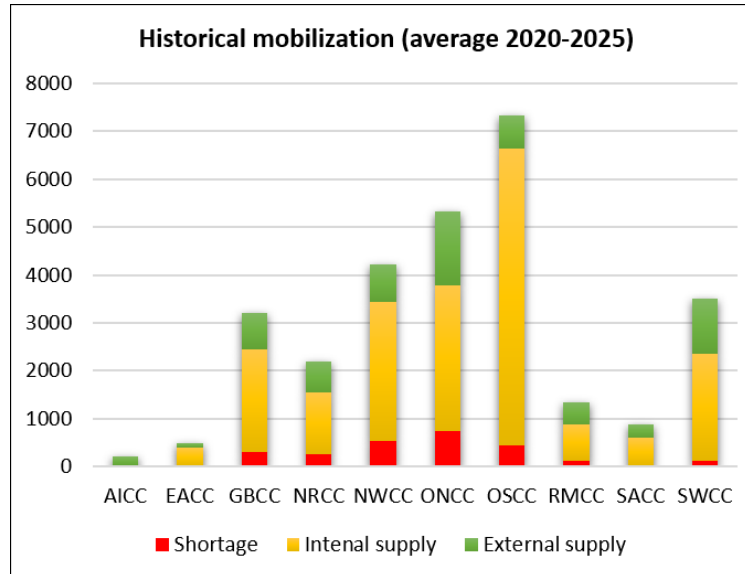
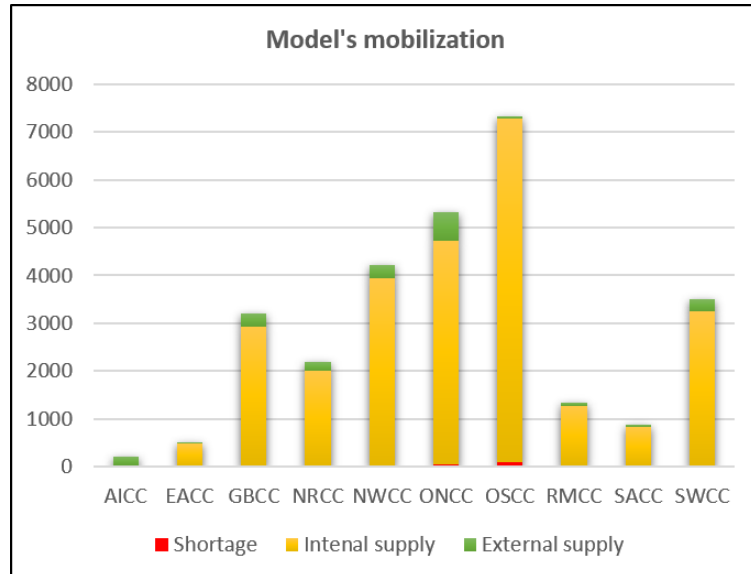
Where:

Demand = Shortage + Supply

Supply = Internal supply + External supply

Internal supply: number of days resources from a GACC are used in that same GACC

External supply: number of days resources other GACCs sent to this GACC to help



3. Results (by stage-2 solutions) vs Historical Data

c, **Resource mobilization:** cherry pick model with 100% staffing levels and GACCs' medium risk-tolerant level (share up to 50% resources)

Model 100% - Medium risk (stage-2)

Crew_1	AICC	NWCC	ONCC	OSCC	NRCC	GBCC	SWCC	RMCC	EACC	SACC
AICC	0	0	0	0	0	0	0	0	0	0
NWCC	88	798	21	0	13	2	1	3	0	0
ONCC	31	87	764	1	16	23	2	0	0	0
OSCC	3	36	123	1,194	1	143	55	0	0	0
NRCC	3	10	7	0	367	4	2	12	1	0
GBCC	0	14	14	0	11	527	34	1	0	0
SWCC	0	3	22	5	15	11	908	11	0	0
RMCC	0	4	2	0	27	3	33	269	5	1
EACC	0	0	0	0	0	0	0	0	23	0
SACC	0	0	1	1	3	1	0	3	5	51

Dozer_2	AICC	NWCC	ONCC	OSCC	NRCC	GBCC	SWCC	RMCC	EACC	SACC
AICC	0	0	0	0	0	0	0	0	0	0
NWCC	0	16	0	0	0	0	0	0	0	0
ONCC	0	3	285	3	0	0	0	0	0	0
OSCC	0	0	3	359	0	15	0	0	0	0
NRCC	0	0	0	0	0	0	0	0	0	0
GBCC	0	0	0	0	0	0	0	0	0	0
SWCC	0	0	1	1	0	3	100	0	0	0
RMCC	0	0	0	0	2	3	0	35	0	0
EACC	0	0	0	0	0	0	0	0	0	0
SACC	0	0	0	0	0	1	0	0	0	17

Engine_3	AICC	NWCC	ONCC	OSCC	NRCC	GBCC	SWCC	RMCC	EACC	SACC
AICC	0	0	0	0	0	0	0	0	0	0
NWCC	1	549	2	0	1	0	0	1	0	0
ONCC	0	4	2,723	0	0	0	0	0	0	0
OSCC	0	0	6	3,762	0	1	0	0	0	0
NRCC	0	0	0	0	0	0	0	0	0	0
GBCC	0	0	0	0	11	411	0	4	0	0
SWCC	0	0	0	0	0	0	654	11	0	4
RMCC	0	0	0	0	39	0	0	56	3	13
EACC	0	0	0	0	0	0	0	0	12	0
SACC	0	0	0	0	0	0	0	0	0	0

Vertical GACCs = home (host resources)

Horizontal GACCs = away (receive resources)

Module_1	AICC	NWCC	ONCC	OSCC	NRCC	GBCC	SWCC	RMCC	EACC	SACC
AICC	0	0	0	0	0	0	0	0	0	0
NWCC	0	45	0	0	0	0	0	0	0	0
ONCC	0	0	26	0	0	0	0	0	0	0
OSCC	0	0	0	0	0	0	0	0	0	0
NRCC	0	0	0	0	39	0	0	0	0	0
GBCC	7	16	5	5	2	120	11	0	0	0
SWCC	0	0	0	0	0	0	0	0	0	0
RMCC	4	5	1	1	10	3	28	78	3	0
EACC	0	0	0	0	0	0	0	0	10	0
SACC	0	0	0	0	0	0	0	0	0	0

Water_2	AICC	NWCC	ONCC	OSCC	NRCC	GBCC	SWCC	RMCC	EACC	SACC
AICC	0	0	0	0	0	0	0	0	0	0
NWCC	0	8	0	0	0	0	0	0	0	0
ONCC	0	5	308	0	0	0	1	0	0	0
OSCC	0	0	0	570	0	19	18	3	0	0
NRCC	0	0	0	0	7	0	0	5	0	0
GBCC	0	0	0	0	0	0	0	0	0	0
SWCC	0	0	0	0	0	0	24	1	0	5
RMCC	0	0	0	0	0	0	0	0	0	0
EACC	0	0	0	0	0	0	0	0	0	0
SACC	0	0	0	0	0	0	0	0	0	0

This is **stage-1 solution** of crew type 1 movement to illustrate why stage 2-is important for mobilization efficiency

Crew_1	AICC	NWCC	ONCC	OSCC	NRCC	GBCC	SWCC	RMCC	EACC	SACC
AICC	0	0	0	0	0	0	0	0	0	0
NWCC	21	394	63	73	38	74	88	23	8	7
ONCC	18	123	483	84	64	89	89	19	1	3
OSCC	40	178	132	817	77	145	158	32	10	14
NRCC	9	54	37	31	153	41	40	11	1	1
GBCC	12	49	51	25	24	241	32	9	0	2
SWCC	20	125	111	74	64	84	517	36	3	7
RMCC	10	35	32	17	15	28	28	171	0	1
EACC	0	0	0	0	0	0	0	0	9	0
SACC	4	16	16	9	8	14	12	6	0	13

Discussions

1. Staffing levels for every GACC: more reasonable/practical numbers suggested by USFS?
2. Predetermine weights for shortages of different resource types? e.g. lacking 1 crew vs lacking 1 engine may not have the same consequence in practice?
3. More realistic resource substitution rule?
4. More detailed resource mobilization & drawdown rules?
5. Other operational restrictions should be captured?
6. Resource uses other than wildfires (prescribed fires, prepositions, others)?
7. ...